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Budget Bites – Project Report

1. Project Summary

Budget Bites is a Flutter mobile application developed to help college students manage their food spending. The app allows users to manage meals, create meal plans, track purchases, and set a budget preference. It also includes Lunch Lens, a rule-based feature that predicts meal prices using previous purchases. The app uses local storage so users can access their saved information without an internet connection.

2. Development Process

I developed Budget Bites in stages instead of trying to build the entire app at once. I started by setting up the Flutter project and implementing the main screens. From there, I added SQLite storage for meals and purchase history, followed by meal planning, budget settings, and the Lunch Lens prediction feature.

I used GitHub branches to organize my work and merged completed feature into the main branch. I also used ClickUp to track tasks and monitor my progress throughout development.

After implementing the main features, I tested the app, fixed issues I encountered, and prepared the project documentation.

3. Challenges and Solutions

During development and testing, I encountered a few issues that needed to be fixed.

Meal Deletion: One issue was that deleting a meal did not remove its associated purchase records. To fix this, I updated the database helper to use a transaction that deletes the related purchase records before deleting the meal itself.

Landscape Layout: During testing, some dialogs had overflow problems when the emulator was rotated into landscape mode. I fixed this by making the affected dialogs scrollable so their content could still be accessed on smaller screen layouts.

Larger Text Sizes: I also noticed that the bottom navigation layout had overflow problems when the device's font size was increased. I adjusted the navigation layout by placing the icons above their labels instead of keeping them side by side.

After making these changes, I tested the affected features again to confirm that the issues were resolved.

4. Testing and Reflection

After implementing the features, I tested Budget Bites on two Android emulators to make sure the app worked as expected. I tested the main features, including meal management, meal planning, purchase history, budget settings, and Lunch Lens. I also checked data persistence after restarting the app, offline functionality, landscape mode, and larger text sizes.

Working on this project gave me more experience with Flutter, SQLite, and managing application data. I also learned how important testing is because some issues, especially the layout problems, were not noticeable until I tested the app under different conditions.

One improvement I would consider in the future is making Lunch Lens more advanced by including additional factors in its predictions instead of relying only on the last three purchases because depending on the situation it can be unreliable or inaccurate at times. Overall, this project helped me better understand how different parts of a mobile application work together that even the in-class activities did miss a bit.